# System of Particles & Rotational Motion

NEET Exam Formula Sheet • Last-Day Revision

## 1 Centre of Mass (COM)

**Definition & Position:**

$$x_{cm} = \frac{m_1x_1 + m_2x_2 + \cdots}{m_1 + m_2 + \cdots} = \frac{\sum m_i x_i}{\sum m_i}$$

$$\vec{r}_{cm} = \frac{\sum m_i \vec{r}_i}{M}$$

**Two-body system:**

$$x_{cm} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

**COM of uniform bodies:**

- Uniform rod: midpoint
- Triangle: centroid $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$
- Semicircular ring: $y_{cm} = \frac{2R}{\pi}$
- Semicircular disc: $y_{cm} = \frac{4R}{3\pi}$
- Solid hemisphere: $y_{cm} = \frac{3R}{8}$
- Hollow hemisphere: $y_{cm} = \frac{R}{2}$

**[Important]** For a system with a cavity: use

$$x_{cm} = \frac{Mx_{full} - m_{cavity}\, x_{cavity}}{M - m_{cavity}}$$

**Motion of COM:**

$$M\vec{a}_{cm} = \vec{F}_{ext}, \quad \vec{v}_{cm} = \frac{\sum m_i \vec{v}_i}{M}$$

**[Key Result]** COM moves as if all external force acts on a point mass $M$ at COM.

## 2 Linear Momentum & Conservation

$$\vec{p}_{total} = M\vec{v}_{cm} = \sum m_i \vec{v}_i$$

$$\vec{F}_{ext} = \frac{d\vec{p}_{total}}{dt}$$

**[Important]** If $\vec{F}_{ext} = 0$, then $\vec{p}_{total} = \text{const.}$

**Rocket propulsion:**

$$F_{thrust} = v_{rel}\frac{dm}{dt}, \quad v = u + v_{rel}\ln\left(\frac{m_0}{m}\right)$$

## 3 Collisions

**Coefficient of restitution:**

$$e = \frac{\text{Relative speed of separation}}{\text{Relative speed of approach}} = \frac{v_2 - v_1}{u_1 - u_2}$$

- Perfectly elastic: $e = 1$
- Perfectly inelastic: $e = 0$
- Partially inelastic: $0 < e < 1$

**1-D Elastic Collision ($m_1, m_2$):**

$$v_1 = \frac{(m_1 - m_2)u_1 + 2m_2u_2}{m_1 + m_2}$$

$$v_2 = \frac{(m_2 - m_1)u_2 + 2m_1u_1}{m_1 + m_2}$$

**[Shortcut]** If $m_1 = m_2$: velocities **exchange**. If $m_1 \gg m_2$: $v_1 \approx u_1$, $v_2 \approx 2u_1 - u_2$.

**Perfectly Inelastic ($u_2 = 0$):**

$$v_{common} = \frac{m_1u_1}{m_1 + m_2}$$

$$\text{Loss in KE} = \frac{1}{2}\cdot\frac{m_1m_2}{m_1 + m_2}(u_1 - u_2)^2$$

**Energy in elastic collision:**

$$\frac{KE_{transferred}}{KE_{initial}} = \frac{4m_1m_2}{(m_1 + m_2)^2}$$

**[Key Result]** Maximum energy transfer when $m_1 = m_2$ (100%).

## 4 Angular Kinematics

**Angular quantities:**

$$\omega = \frac{d\theta}{dt}, \quad \alpha = \frac{d\omega}{dt}$$

**Equations of motion (const. $\alpha$):**

$$\omega = \omega_0 + \alpha t$$
$$\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2$$
$$\omega^2 = \omega_0^2 + 2\alpha\theta$$
$$\theta = \tfrac{1}{2}(\omega_0 + \omega)t$$

**[Shortcut]** Angle in $n^{th}$ second: $\theta_n = \omega_0 + \frac{\alpha}{2}(2n - 1)$

## 5 Linear ↔ Angular Relations

$$v = r\omega \qquad a_t = r\alpha$$

$$a_c = r\omega^2 = \frac{v^2}{r}$$

$$s = r\theta$$

**[Key Result]** $v$, $a_t$, $s$ are tangential (along rim); $a_c$ is centripetal (inward).

## 6 Torque & Rotational Dynamics

**Torque:**

$$\vec{\tau} = \vec{r} \times \vec{F}, \quad |\tau| = rF\sin\theta = F \cdot d$$

where $d$ = perpendicular distance (moment arm).

**Newton's 2nd Law (rotation):**

$$\boxed{\tau_{net} = I\alpha}$$

**Equilibrium conditions:**

$$\sum \vec{F} = 0 \quad \text{(translational)}, \quad \sum \vec{\tau} = 0 \quad \text{(rotational)}$$

**[Important]** Couple: two equal, opposite, non-collinear forces. Torque = $F \times d$ (independent of pivot).

## 7 Moment of Inertia (MI)

**Definition:**

$$I = \sum m_i r_i^2 = \int r^2\, dm$$

**Radius of gyration:**

$$I = MK^2 \implies K = \sqrt{I/M}$$

**Standard Results:**

| Body | Axis | $I$ |
|---|---|---|
| Thin rod ($L$) | Centre $\perp$ | $\frac{ML^2}{12}$ |
| Thin rod ($L$) | End $\perp$ | $\frac{ML^2}{3}$ |
| Solid disc ($R$) | Central $\perp$ | $\frac{MR^2}{2}$ |
| Solid disc ($R$) | Diameter | $\frac{MR^2}{4}$ |
| Circular ring ($R$) | Central $\perp$ | $MR^2$ |
| Circular ring ($R$) | Diameter | $\frac{MR^2}{2}$ |
| Solid sphere ($R$) | Diameter | $\frac{2MR^2}{5}$ |
| Hollow sphere ($R$) | Diameter | $\frac{2MR^2}{3}$ |
| Solid cylinder ($R$) | Own axis | $\frac{MR^2}{2}$ |
| Solid cylinder ($R$) | Transverse | $\frac{M}{12}(3R^2+L^2)$ |
| Hollow cylinder ($R$) | Own axis | $MR^2$ |
| Rectangular plate | Central $\perp$ | $\frac{M(a^2+b^2)}{12}$ |

**Theorems:**
**[Important]** Parallel Axis Theorem:

$$I = I_{cm} + Md^2$$

($d$ = distance between axes, both parallel)
**[Important]** Perpendicular Axis Theorem (Plane lamina only):

$$I_z = I_x + I_y$$

## 8 Angular Momentum & Conservation

**For a particle:**

$$\vec{L} = \vec{r} \times \vec{p} = m(\vec{r} \times \vec{v}), \quad |\vec{L}| = mvr\sin\theta$$

**For a rigid body:**

$$L = I\omega, \quad \vec{\tau} = \frac{d\vec{L}}{dt}$$

**[Important]** If $\tau_{ext} = 0$, then $L = I\omega = \text{const.}$
**[Shortcut]** $I_1\omega_1 = I_2\omega_2$ — used in figure skater / diver problems.

**Relation between $L$ and $\tau$:**

$$\tau = \frac{\Delta L}{\Delta t}$$

## 9 Work, Energy & Power (Rotation)

**Rotational KE:**

$$KE_{rot} = \frac{1}{2}I\omega^2 = \frac{L^2}{2I} = \frac{(\tau t)^2}{2I}$$

**Work done by torque:**

$$W = \tau \cdot \theta \quad (\text{const. } \tau), \quad W = \int \tau \, d\theta$$

**Work-Energy theorem:**

$$W_{net} = \Delta KE_{rot} = \frac{1}{2}I\omega_f^2 - \frac{1}{2}I\omega_i^2$$

**Power:**

$$P = \tau\omega$$

## 10 Rolling Motion (Pure Rolling)

**[Important]** Condition for pure rolling (no slip):

$$v_{cm} = R\omega$$

**Total KE of rolling body:**

$$KE_{total} = \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2 = \frac{1}{2}Mv_{cm}^2\left(1 + \frac{K^2}{R^2}\right)$$

**[Shortcut]** $K^2/R^2$: solid sphere $= \frac{2}{5}$, solid cylinder $= \frac{1}{2}$, hollow sphere $= \frac{2}{3}$, ring/hollow cyl $= 1$
**Rolling on incline (angle $\theta$):**

$$a = \frac{g\sin\theta}{1 + K^2/R^2}$$

$$v_{bottom} = \sqrt{\frac{2gh}{1 + K^2/R^2}}$$

**[Key Result]** More $K^2/R^2 \Rightarrow$ slower acceleration. Solid sphere rolls fastest; ring rolls slowest.
**Friction in rolling:** Static friction acts; no energy loss.
**Instantaneous axis (pure rolling):** Bottom contact point ($v = 0$).

$$v_{top} = 2v_{cm}, \quad v_{cm(\text{point})} = v_{cm}$$

# Most Important Formulas at a Glance

| Topic | Formula | Remarks |
|---|---|---|
| COM (2 bodies) | $x_{cm} = \dfrac{m_1x_1 + m_2x_2}{m_1 + m_2}$ | Generalise to 3D with $y_{cm}$, $z_{cm}$ |
| Semicircular ring COM | $y_{cm} = \dfrac{2R}{\pi}$ | From centre/diameter |
| Elastic collision speeds | $v_1 = \dfrac{(m_1 - m_2)u_1 + 2m_2u_2}{m_1 + m_2}$ | Both momentum & KE conserved |
| Coeff. of restitution | $e = \dfrac{v_2 - v_1}{u_1 - u_2}$ | $e = 1$ elastic; $e = 0$ perfectly inelastic |
| Torque | $\tau = rF\sin\theta = I\alpha$ | $\tau = 0 \Rightarrow L = \text{const}$ |
| MI (Parallel Axis) | $I = I_{cm} + Md^2$ | Any axis $\parallel$ to CM axis |
| MI (Perp. Axis) | $I_z = I_x + I_y$ | Plane laminae only |
| Angular momentum | $L = I\omega = mvr\sin\theta$ | Conserved when $\tau_{ext} = 0$ |
| Rotational KE | $KE = \dfrac{1}{2}I\omega^2$ | Analogous to $\frac{1}{2}mv^2$ |
| Rolling acceleration | $a = \dfrac{g\sin\theta}{1 + K^2/R^2}$ | Largest for solid sphere ($K^2/R^2 = \frac{2}{5}$) |
| Rolling KE (total) | $KE = \dfrac{1}{2}Mv_{cm}^2\left(1 + \dfrac{K^2}{R^2}\right)$ | Sum of translational + rotational |
| Pure rolling condition | $v_{cm} = R\omega$ | No slipping at contact point |
| Power (rotation) | $P = \tau\omega$ | Analogous to $P = Fv$ |
| Work by torque | $W = \tau\theta$ | For constant torque |
| Angular kinematics | $\omega^2 = \omega_0^2 + 2\alpha\theta$ | Direct analogue of $v^2 = u^2 + 2as$ |

**EXAM TIPS**

- In COM problems with absent part, subtract the removed portion.
- Always check if $\tau_{ext} = 0$ before applying $L$-conservation.
- Perp. Axis theorem: ONLY for flat laminae (2D objects).
- Rolling without slipping $\Rightarrow$ static friction (does no work).

**$K^2/R^2$ QUICK TABLE**

| | |
|---|---|
| Solid sphere | 2/5 |
| Solid cylinder | 1/2 |
| Hollow sphere | 2/3 |
| Ring / hollow cyl | 1 |
| Disc (diametric) | 1/4 |

**COM OF STANDARD SHAPES**

| | |
|---|---|
| Semicircle ring | $2R/\pi$ |
| Semicircle disc | $4R/3\pi$ |
| Solid hemisphere | $3R/8$ |
| Hollow hemisphere | $R/2$ |
| Cone (solid) | $h/4$ from base |

All standard SI units assumed. $M$ = total mass, $R$ = radius, $L$ = length, $h$ = height, $K$ = radius of gyration.